

CONTACT

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shehanjamal.github.io/github-portfolio

github.com/shehanjamal

Grafton, Auckland, New Zealand

KEY SKILLS

LANGUAGES: Python, TypeScript, JavaScript, Go, C#, SQL, PL/SQL

FRONTEND: React, Next.js, Leaflet, Electron, HTML/CSS

BACKEND AND APIS: FastAPI, .NET, Node.js, REST, GraphQL, Apollo, OpenAPI 3.0, Auth0

DATABASES: PostgreSQL, PostGIS, MySQL, Supabase

CLOUD AND DEVOPS: AWS, Azure, Docker, GitHub Actions, CI/CD, Git

MACHINE LEARNING: CatBoost, scikit-learn, TreeSHAP, BERT, LSTM, OpenCV, Tesseract OCR

TOOLS: Jira, GitHub, Postman, Visual Studio Code

CORE COMPETENCIES: System design, API design, technical documentation, client advisory, cross-team collaboration

EDUCATION

MASTER OF SOFTWARE ENGINEERING

Yoobee College of Creative Innovation, New Zealand, 2025 – 2026

BENG (HONS) SOFTWARE ENGINEERING WITH INDUSTRIAL PLACEMENT

Informatics Institute of Technology, Sri Lanka (University of Westminster, UK), 2018 – 2021

CERTIFICATES

AWS Cloud Foundation (AWS Academy)

AWS Cloud Security Fundamentals (AWS Academy)

Cutting Edge 2018: 1st Place (Project) and 1st Place (Creativity)

PUBLICATION

“A Literature Review of Recommendation Systems”, 2020 IEEE International Conference for Innovation and Technology (INOCON)

SHEHAN ROSHAN JAMAL

Software Engineer

PROFILE

Software engineer with a Master of Software Engineering and four years of experience building and delivering full-stack systems for enterprise and research clients across Poland, the Netherlands, Australia, the US and New Zealand. Owned a geospatial research platform end to end, from requirements analysis and schema design through to production, and built an explainable-AI prediction system achieving 92% accuracy. Strong across Python, TypeScript, React/Next.js, PostgreSQL and AWS, with C#/.NET experience from enterprise ERP work, and a track record of taking systems from design documentation through to deployment.

PROFESSIONAL EXPERIENCE

TIAKI TAONGA TRUST | April 2026 – July 2026

Software Engineer (Internship), Auckland, New Zealand

Portfolio Owner: Geo-Spatial & Policy Research Platform

A platform for mapping and managing cultural and policy research data, built end to end. Tech: TypeScript, React, Next.js, PostgreSQL, PostGIS, AWS, Supabase, Leaflet, GitHub Actions, Jira.

- Owned the platform end to end as Portfolio Owner, taking it from requirements analysis through database schema design, API design and implementation to production deployment.
- Designed a PostgreSQL/PostGIS schema on a hub-and-spoke model that made orphaned records structurally impossible, enforcing integrity through foreign-key and CHECK constraints and audit triggers that produced an immutable change log.
- Built 25+ REST endpoints in TypeScript with fully parameterised SQL, a standardised response envelope, custom domain error codes and pagination, documented in a published OpenAPI 3.0 specification.
- Built the React/Next.js frontend with Leaflet interactive mapping and a 7-role RBAC system, extended by a delegated per-project permission model enabling granular access control without code changes.
- Led cross-portfolio authentication integration into a shared monorepo, resolving an identity mismatch between two independently built systems while leaving all 25+ downstream endpoints unmodified.
- Migrated 19 API routes from Supabase PostgREST to raw parameterised SQL on AWS PostgreSQL, diagnosing spatial-function resolution, geometry serialisation and connection-pooling defects during the cutover.
- Presented live progress demonstrations throughout the engagement and a final end-to-end system demonstration to the Trust's technical and business stakeholders.
- Established CI/CD with GitHub Actions and versioned idempotent SQL migrations delivered via pull request, and produced full handover documentation.

IFS | July 2022 – April 2025

Software Engineer, Sri Lanka

Enterprise ERP delivery for international customers across Poland, the Netherlands, Australia and the US. Tech: PL/SQL, SQL, C#/.NET, XML, Azure, Jira.

- Delivered customisations and new functionality for enterprise customers, working across IFS versions from App09 through Cloud 24R1.
- Advised Polish and Australian customers on PSO and related modules as functional advisor, recommending alternative approaches that reduced the build effort of delivered customisations.
- Managed end-to-end version uplifts (App09 to App10, App10 to Cloud), covering the code, configuration and data changes each migration required.
- Developed across the application stack, covering PSO, dialogs, assistants and custom pages, including C# client extensions built against the IFS .NET Client SDK (App09/App10).
- Built customer-specific reports from scratch and modified existing XML report definitions to introduce new fields and behaviour for individual customer requirements.
- Delivered application localisation into Polish and French, and authored technical specifications and time-estimation sheets used by other developers for project planning.

- Built and remediated Tar scripts for environment data setup, adapting legacy scripts to newer application versions where required data structures had changed.
- Ran internal knowledge-sharing sessions onboarding new team members onto core application functionality.

AGILE FINANCIALS | August 2020 – June 2021

Software Engineer (Internship), Sri Lanka

Tech: Go, React, GraphQL, PostgreSQL, Auth0.

- Built the full sign-up and login flow end to end, covering the React frontend, the Go backend logic and the supporting database records.
- Integrated Auth0 across both the frontend and backend for authentication, and implemented the email confirmation process on top of it.
- Designed and created the PostgreSQL database, and developed the connection layer between the database and the backend API in Go.
- Built the GraphQL API layer and wrote both integration and unit test coverage for it.
- Wrote integration and unit tests for the Go backend, covering the authentication and sign-up paths.
- Contacted the Colombo Stock Exchange directly to source historical market data required for the project's model.
- Presented a full project demonstration to the board members of Nations Trust Bank.

K E Y P R O J E C T S

TFT-PREDICT V3 | 2026

Explainable AI Prediction System, Master's Capstone

Tech: Python, CatBoost, TreeSHAP, FastAPI, Electron, OpenCV, Tesseract OCR.

- Built a real-time prediction system achieving 92.08% Top-4 accuracy and 0.9798 AUC, using a screenshot-only pipeline that fed OCR-extracted game state through 23 engineered features into a CatBoost model.
- Raised trait recognition accuracy from 19.8% to 61.0% by diagnosing and fixing an identifier mismatch between display names and internal game data IDs.
- Delivered model transparency via TreeSHAP, generating human-readable explanations for every prediction and surfacing them through a FastAPI backend and Electron desktop overlay.

REACHOUT | 2025

Volunteer Coordination Platform, Personal Project

Tech: Python, React, Node.js, Apollo, MySQL, Docker, OpenAI API, AssemblyAI.

- Cut task creation time from 3 minutes to 30 seconds by integrating OpenAI and AssemblyAI to provide voice-to-text task entry.
- Architected and containerised the full stack (Python backend, React frontend, Apollo middleware) and deployed it with Docker.

NOVEL-TO-COMIC GENERATION PIPELINE | 2026

Personal Project

Tech: FLUX.1, PuLID, Llama 3.1, ComfyUI, Vast.ai.

- Built an end-to-end pipeline converting novel text into illustrated comic panels on GPU cloud infrastructure, combining FLUX.1 Dev for image generation, PuLID Flux II for character consistency and Llama 3.1 8B for scene parsing.

EMOTION ANALYZER FOR ARTICLES | 2022 – 2023

Research Project

Tech: Python, BERT, LSTM, sentiment analysis, Chrome extension APIs.

- Designed and developed a Google Chrome plugin that analyses the article on the page and rates it on an emotional scale, showing how positive or negative the writing is.
- Built a hybrid model combining a BERT transformer, sentiment analysis and an LSTM to produce the emotional score.
- Developed the Python API that served the model to the browser plugin.

FOOD POINT | 2019 – 2020

Recipe Recommendation System

Tech: Python, K-means clustering, React, Next.js, Node.js, MongoDB.

- Built a recommendation system where users enter the ingredients they have and the model suggests recipes they can make.
- Applied K-means clustering to group recipes by ingredient similarity, and added a review and rating system to surface the most popular results.